

Problem Set 10 (40 pts)

Due at 11:59pm

Friday, November 14

Exercises are from <https://aimacode.github.io/aima-exercises/> after some rewriting and fixing of typos in their original descriptions.

- 1.(12 pts)** (Exercise 13.30) In our analysis of the wumpus world, the agent has explored three squares [1,1], [1,2], and [2, 1] with the following percepts of the breeze condition:

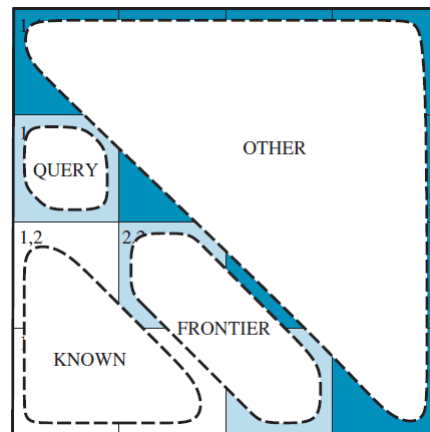
$$b = \neg b_{1,1} \wedge b_{1,2} \wedge b_{2,1}$$

and the knowledge about none of them containing a pit:

$$known = \neg p_{1,1} \wedge \neg p_{1,2} \wedge \neg p_{2,1}$$

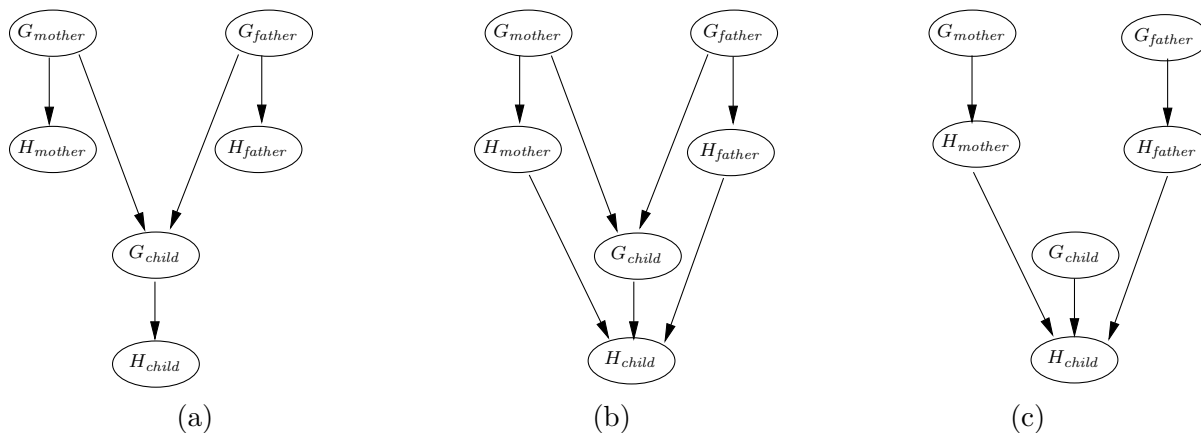
We now assume that each square contains a pit with probability 0.01 independent of the other squares. Calculate the probability distributions $\mathbf{P}(P_{1,3} \mid known, b)$ and $\mathbf{P}(P_{2,2} \mid known, b)$ for pits in [1,3] and [2,2], respectively. What can you say about the relative performance of a logical versus a probabilistic agent in this case (assume that a logical agent makes a random choice)?

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 B OK	2,2	3,2	4,2
1,1 OK	2,1 B OK	3,1	4,1



- 2.(18 pts)** (Exercise 14.7) Let H_x be a random variable denoting the handedness of an individual x , with possible values l or r . A common hypothesis is that left- or right-handedness is inherited by a simple mechanism; that is, perhaps there is a gene G_x , also with values l or r , and perhaps actual handedness turns out mostly the same (with some probability s) as the gene an individual possesses.

Furthermore, perhaps the gene itself is equally likely to be inherited from either of an individual's parents, with a small nonzero probability m of a random mutation flipping the handedness.



(a) (2 pts) Which of the three networks in the above figure claim that

$$P(G_{father}, G_{mother}, G_{child}) = P(G_{father})P(G_{mother})P(G_{child})?$$

(b) (3 pts) Which of the three networks make independence claims that are consistent with the hypothesis about the inheritance of handedness?

(c) (3 pts) Which of the three networks is the best description of the hypothesis?

(d) (4 pts) Write down the CPT for the G_{child} node in network (a), in terms of s and m .

(e) (6 pts) Suppose that $P(G_{father} = l) = P(G_{mother} = l) = q$. In network (a), derive an expression for $P(G_{child} = l)$ in terms of m and q only, by conditioning on its parent nodes.

3.(10 pts) (Exercise 14.13) In your local nuclear power station, there is an alarm that senses when a temperature gauge exceeds a given threshold. The gauge measures the temperature of the core. Consider the Boolean variables A (alarm sounds), F_A (alarm is faulty), and F_G (gauge is faulty) and the multivalued nodes G (gauge reading) and T (actual core temperature).

(a) (6 pts) Draw a Bayesian network for this domain, given that the gauge is more likely to fail when the core temperature gets too high.

(c) (4 pts) Suppose there are just two possible actual and measured temperatures, normal and high; the probability that the gauge gives the correct temperature is x when it is working, but y when it is faulty. Give the conditional probability table associated with G .